

Advanced Database Management Systems (CSC-234)

JSON Data

JavaScript Object Notation (JSON)

- Another data model, like XML and relational data
- Standard for “serializing” data objects usually in files
- Human-readable, useful for data interchange
- Also useful for representing & storing semi-structured data

```
{ "Books":  
  [  
    { "ISBN":"ISBN-0-13-713526-2",  
      "Price":85,  
      "Edition":3,  
      "Title":"A First Course in Database Systems",  
      "Authors":[ {"First_Name":"Jeffrey", "Last_Name":"Ullman"},  
                   {"First_Name":"Jennifer", "Last_Name":"Widom"} ] }  
    ,  
    { "ISBN":"ISBN-0-13-815504-6",  
      "Price":100,  
      "Remark":"Buy this book bundled with 'A First Course' - a great deal!",  
      "Title":"Database Systems:The Complete Book",  
      "Authors":[ {"First_Name":"Hector", "Last_Name":"Garcia-Molina"},  
                   {"First_Name":"Jeffrey", "Last_Name":"Ullman"},  
                   {"First_Name":"Jennifer", "Last_Name":"Widom"} ] }  
  ]  
}
```

JavaScript Object Notation (JSON)

- No longer tied to JavaScript

- Parsers for many languages

```
{ "Books":  
  { "ISBN":"ISBN-0-13-713526-2",  
    "Price":85,  
    "Edition":3,  
    "Title":"A First Course in Database Systems",  
    "Authors":[ {"First_Name":"Jeffrey", "Last_Name":"Ullman"},  
                 {"First_Name":"Jennifer", "Last_Name":"Widom"} ] }  
  ,  
  { "ISBN":"ISBN-0-13-815504-6",  
    "Price":100,  
    "Remark":"Buy this book bundled with 'A First Course' - a great deal!",  
    "Title":"Database Systems:The Complete Book",  
    "Authors":[ {"First_Name":"Hector", "Last_Name":"Garcia-Molina"},  
                 {"First_Name":"Jeffrey", "Last_Name":"Ullman"},  
                 {"First_Name":"Jennifer", "Last_Name":"Widom"} ] }  
}
```

JSON

- Basic constructs (recursive)

- Base values

- Number, string, boolean

- Objects { }

- Set of label-value pairs

- Arrays []

- List of values

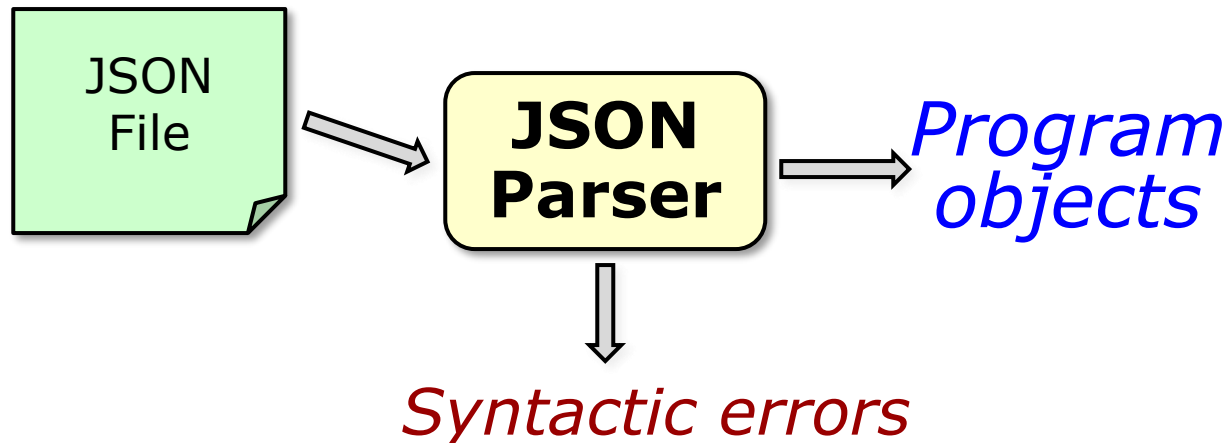
```
{ "Books":  
  [  
    { "ISBN":"ISBN-0-13-713526-2",  
      "Price":85,  
      "Edition":3,  
      "Title":"A First Course in Database Systems",  
      "Authors":[ { "First_Name":"Jeffrey", "Last_Name":"Ullman"},  
                   { "First_Name":"Jennifer", "Last_Name":"Widom"} ] }  
    ,  
    { "ISBN":"ISBN-0-13-815504-6",  
      "Price":100,  
      "Remark":"Buy this book bundled with 'A First Course' - a great deal!",  
      "Title":"Database Systems:The Complete Book",  
      "Authors":[ { "First_Name":"Hector", "Last_Name":"Garcia-Molina"},  
                   { "First_Name":"Jeffrey", "Last_Name":"Ullman"},  
                   { "First_Name":"Jennifer", "Last_Name":"Widom"} ] }  
  ],  
  "Magazines":  
  [  
    { "Title":"National Geographic",  
      "Month":"January",  
      "Year":2009 }  
    ,  
    { "Title":"Newsweek",  
      "Month":"February",  
      "Year":2009 }  
  ]  
}
```

Relational Model vs. JSON

	Relational	JSON
Structure	Tables	Nested sets and arrays
Schema	Fixed in advance	Self-describing, flexible
Queries	Simple, expressive languages	Nothing that is as widely used
Ordering	None	Arrays
Implementation	Native systems	Coupled with programming languages; NoSQL systems

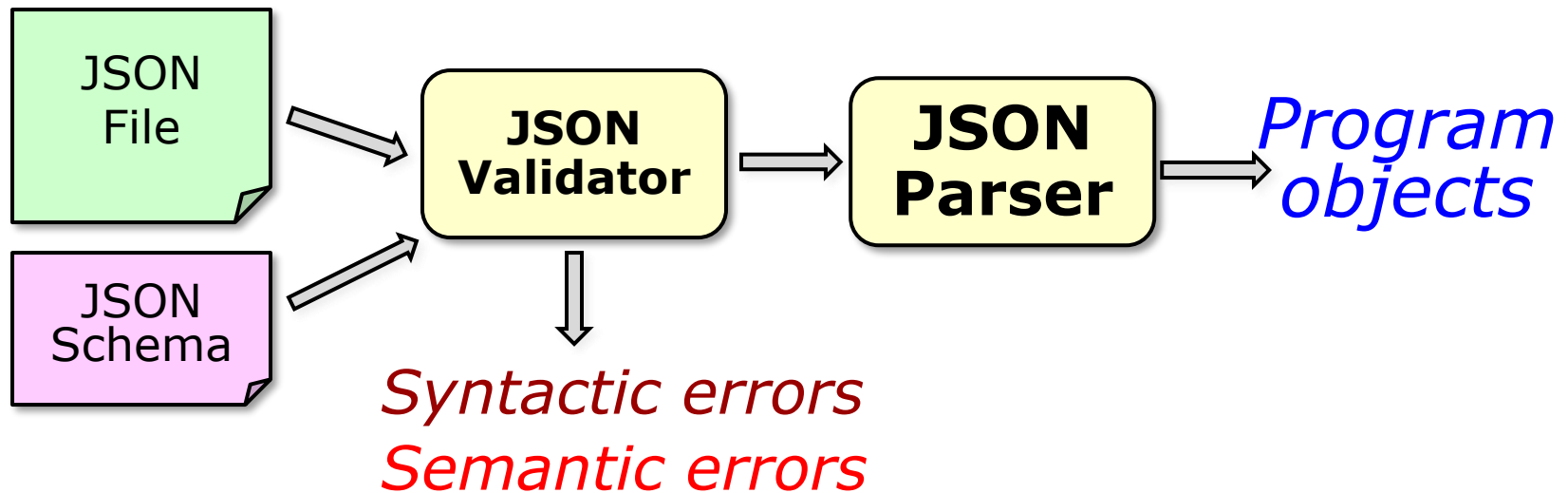
Syntactically Valid JSON

- Adheres to basic structural requirements
 - Sets of label-value pairs
 - Arrays of values
 - Base values from predefined types



Semantically Valid JSON

- Adheres to basic structural requirements + conforms to specified schema



Demo

- Constructs and syntactic correctness
- Flexibility
- JSON schema, validation

```
{ "Books":  
  [  
    { "ISBN":"ISBN-0-13-713526-2",  
      "Price":85,  
      "Edition":3,  
      "Title":"A First Course in Database Systems",  
      "Authors":[ {"First_Name":"Jeffrey", "Last_Name":"Ullman"},  
                   {"First_Name":"Jennifer", "Last_Name":"Widom"} ] }  
    ,  
    { "ISBN":"ISBN-0-13-815504-6",  
      "Price":100,  
      "Remark":"Buy this book bundled with 'A First Course' - a great deal!",  
      "Title":"Database Systems:The Complete Book",  
      "Authors":[ {"First_Name":"Hector", "Last_Name":"Garcia-Molina"},  
                   {"First_Name":"Jeffrey", "Last_Name":"Ullman"},  
                   {"First_Name":"Jennifer", "Last_Name":"Widom"} ] }  
  ]  
}
```

Demo Data (Bookstore.json)

```
{ "Books":  
  [  
    { "ISBN":"ISBN-0-13-713526-2",  
      "Price":85,  
      "Edition":3,  
      "Title":"A First Course in Database Systems",  
      "Authors":[ {"First_Name":"Jeffrey", "Last_Name":"Ullman"},  
                   {"First_Name":"Jennifer", "Last_Name":"Widom"} ] },  
    { "ISBN":"ISBN-0-13-815504-6",  
      "Price":100,  
      "Remark":"Buy this book bundled with 'A First Course' - a great deal!",  
      "Title":"Database Systems:The Complete Book",  
      "Authors":[ {"First_Name":"Hector", "Last_Name":"Garcia-Molina"},  
                   {"First_Name":"Jeffrey", "Last_Name":"Ullman"},  
                   {"First_Name":"Jennifer", "Last_Name":"Widom"} ] }  
  ],  
  "Magazines":  
  [  
    { "Title":"National Geographic",  
      "Month":"January",  
      "Year":2009 },  
    { "Title":"Newsweek",  
      "Month":"February",  
      "Year":2009 }  
  ]  
}
```

powered by ace

```
1 { "Books":
2   [
3     { "ISBN":"ISBN-0-13-713526-2",
4       "Price":85,
5       "Edition":3,
6       "Title":"A First Course in Database Systems",
7       "Authors":[ {"First_Name":"Jeffrey", "Last_Name":"Ullman"},
8                   {"First_Name":"Jennifer", "Last_Name":"Widom"} ]
9     },
10    { "ISBN":"ISBN-0-13-815504-6",
11      "Price":100,
12      "Remark":"Buy this book bundled with 'A First Course' - a
13      great deal!",
14      "Title":"Database Systems:The Complete Book",
15      "Authors":[ {"First_Name":"Hector", "Last_Name":"Garcia
16                  -Molina"},
17                  {"First_Name":"Jeffrey", "Last_Name":"Ullman"},
18                  {"First_Name":"Jennifer", "Last_Name":"Widom"} ]
19    },
20  ],
21  "Magazines":
22  [
23    { "Title":"National Geographic",
24      "Month":"January",
25      "Year":2009 }
26  ],
27  { "Title":"Newsweek",
28    "Month":"February",
```

object {2}

- Books [2]
 - 0 {5}
 - ISBN : ISBN-0-13-713526-2
 - Price : 85
 - Edition : 3
 - Title : A First Course in Database Systems
 - Authors [2]
 - 1 {5}
 - Magazines [2]
 - 0 {3}
 - Title : National Geographic
 - Month : January
 - Year : 2009
 - 1 {3}
 - Title : Newsweek
 - Month : February
 - Year : 2009



```
1 { "Books":  
2   [  
3     { "ISBN":"ISBN-0-13-713526-2",  
4       "Price":85  
5       "Edition":3,  
6       "Title":"A First Course in Database Systems",  
7       "Authors":[ {"First_Name":"Jeffrey", "Last_Name":"Ullman"},  
8                   {"First_Name":"Jennifer", "Last_Name":"Widom"} ]  
9     },  
10    { "ISBN":"ISBN-0-13-815504-6",  
11      "Price":100,  
12      "Remark":"Buy this book bundled with 'A First Course' - a  
13      great deal!",  
14      "Title":"Database Systems:The Complete Book",  
15      "Authors":[ {"First_Name":"Hector", "Last_Name":"Garcia  
16                  -Molina"},  
17                  {"First_Name":"Jeffrey", "Last_Name":"Ullman"},  
18                  {"First_Name":"Jennifer", "Last_Name":"Widom"} ]  
19    },  
20    ],  
21    "Magazines":  
22    [  
23      { "Title":"National Geographic",  
24        "Month":"January",  
25        "Year":2009 }  
26      ,  
27      { "Title":"Newsweek",  
28        "Month":"February",
```

Error?

COMMA MISSING

powered by ace

```
1 { "Books":
2   [
3     { "ISBN":"ISBN-0-13-713526-2",
4       "Price":85,
5       "Edition":3,
6       "Title":"A First Course in Database Systems",
7       "Authors":[ {"First_Name":"Jeffrey", "Last_Name":"Ullman"}],
8                 {"First_Name":"Jennifer", "Last_Name":"Widom"} ]
9     },
10    { "ISBN":"ISBN-0-13-815504-6",
11      "Price":100,
12      "Remark":"Buy this book bundled with 'A First Course' - a
13      great deal!",
14      "Title":"Database Systems:The Complete Book",
15      "Authors":[ {"First_Name":"Hector", "Last_Name":"Garcia
16                  -Molina"},
17                  {"First_Name":"Jeffrey", "Last_Name":"Ullman"},
18                  {"First_Name":"Jennifer", "Last_Name":"Widom"} ]
19    },
20    ],
21    "Magazines":
22    [
23      { "Title":"National Geographic",
24        "Month":"January",
25        "Year":2009 }
26      ,
27      { "Title":"Newsweek",
28        "Month":"February",
```

Error?

EXTRA BRACE

powered by ace

```
1 { "Books":  
2  [  
3    { "ISBN":ISBN-0-13-713526-2,  
4      "Price":85,  
5      "Edition":3,  
6      "Title":"A First Course in Database Systems",  
7      "Authors":[ {"First_Name":"Jeffrey", "Last_Name":"Ullman"},  
8                  {"First_Name":"Jennifer", "Last_Name":"Widom"} ]  
9    },  
10   { "ISBN":"ISBN-0-13-815504-6",  
11     "Price":100,  
12     "Remark":"Buy this book bundled with 'A First Course' - a  
13     great deal!",  
14     "Title":"Database Systems:The Complete Book",  
15     "Authors":[ {"First_Name":"Hector", "Last_Name":"Garcia  
16                  -Molina"},  
17                  {"First_Name":"Jeffrey", "Last_Name":"Ullman"},  
18                  {"First_Name":"Jennifer", "Last_Name":"Widom"} ]  
19   },  
20   "Magazines":  
21   [  
22     { "Title":"National Geographic",  
23       "Month":"January",  
24       "Year":2009 }  
25     ,  
26     { "Title":"Newsweek",  
27       "Month":"February",
```

Error?

**QUOTES AROUND STRING
MISSING**

Only unquoted values

- Numbers
- null
- true
- false

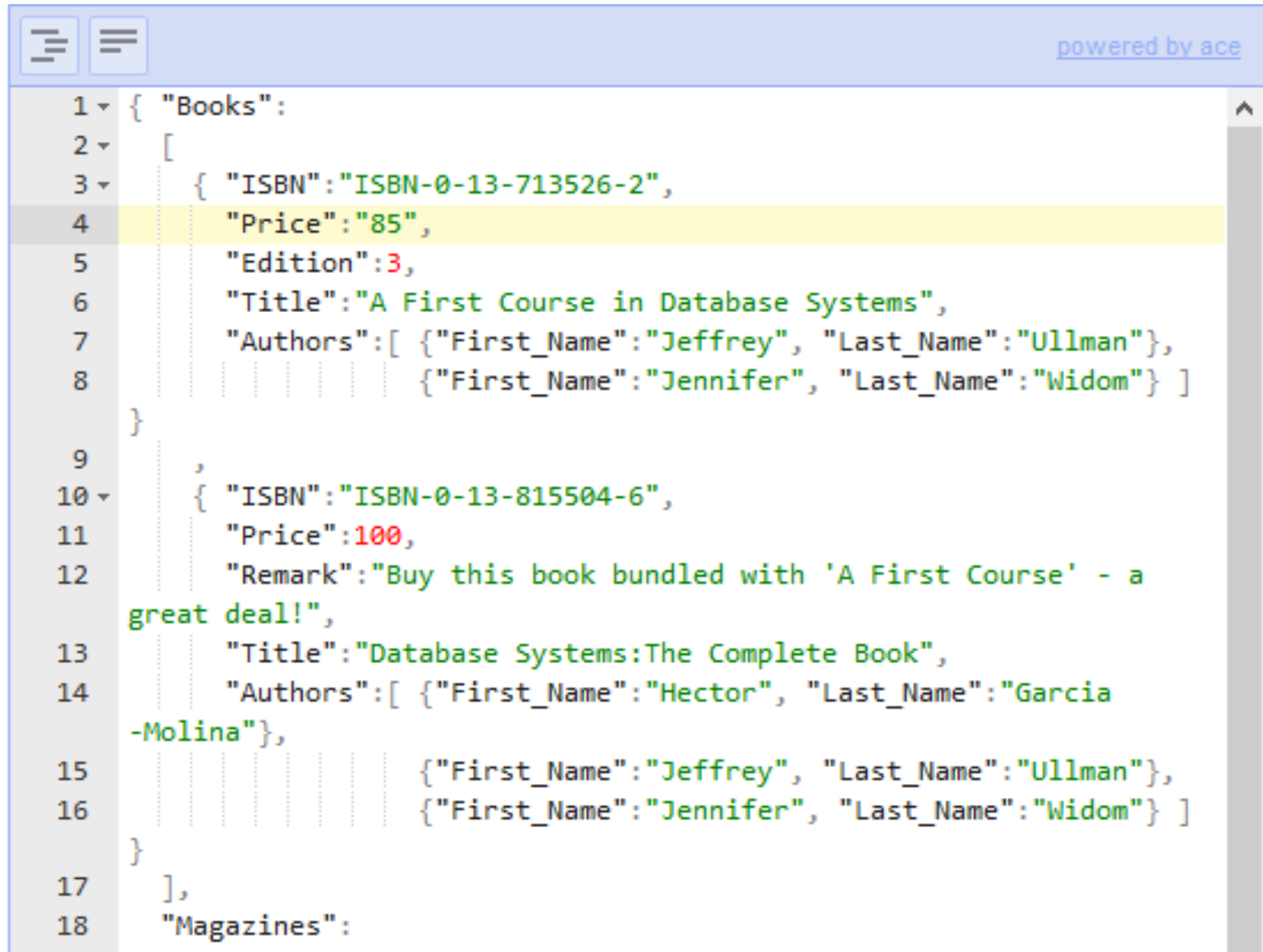


```
1 { "Books":
2   [
3     { "ISBN":"ISBN-0-13-713526-2",
4       Price:85,
5       "Edition":3,
6       "Title":"A First Course in Database Systems",
7       "Authors":[ {"First_Name":"Jeffrey", "Last_Name":"Ullman"},
8                   {"First_Name":"Jennifer", "Last_Name":"Widom"} ]
9     },
10    { "ISBN":"ISBN-0-13-815504-6",
11      "Price":100,
12      "Remark":"Buy this book bundled with 'A First Course' - a
13      great deal!",
14      "Title":"Database Systems:The Complete Book",
15      "Authors":[ {"First_Name":"Hector", "Last_Name":"Garcia
16                  -Molina"},
17                  {"First_Name":"Jeffrey", "Last_Name":"Ullman"},
18                  {"First_Name":"Jennifer", "Last_Name":"Widom"} ]
19    },
20    "Magazines":
21    [
22      { "Title":"National Geographic",
23        "Month":"January",
24        "Year":2009 }
25      ,
26      { "Title":"Newsweek",
27        "Month":"February",
```

Error?

**QUOTES AROUND LABEL
MISSING**

Price value quoted – not an error



```
1 { "Books":  
2   [  
3     { "ISBN":"ISBN-0-13-713526-2",  
4       "Price":"85",  
5       "Edition":3,  
6       "Title":"A First Course in Database Systems",  
7       "Authors":[ {"First_Name":"Jeffrey", "Last_Name":"Ullman"},  
8                   {"First_Name":"Jennifer", "Last_Name":"Widom"} ]  
9     },  
10    { "ISBN":"ISBN-0-13-815504-6",  
11      "Price":100,  
12      "Remark":"Buy this book bundled with 'A First Course' - a  
13      great deal!",  
14      "Title":"Database Systems:The Complete Book",  
15      "Authors":[ {"First_Name":"Hector", "Last_Name":"Garcia  
16                  -Molina"},  
17                  {"First_Name":"Jeffrey", "Last_Name":"Ullman"},  
18                  {"First_Name":"Jennifer", "Last_Name":"Widom"} ]  
19    ],  
20    "Magazines":
```

Null (reserved word) value is acceptable, no quotes

```

1 { "Books":
2   [
3     { "ISBN": "ISBN-0-13-713526-2",
4       "Price": "85",
5       "Edition": null,
6       "Title": "A First Course in Database Systems",
7       "Authors": [ { "First_Name": "Jeffrey", "Last_Name": "Ullman" },
8                   { "First_Name": "Jennifer", "Last_Name": "Widom" } ]
9     },
10    { "ISBN": "ISBN-0-13-815504-6",
11      "Price": 100,
12      "Remark": "Buy this book bundled with 'A First Course' - a
13 great deal!",
14      "Title": "Database Systems: The Complete Book",
15      "Authors": [ { "First_Name": "Hector", "Last_Name": "Garcia
16 -Molina" },
17                  { "First_Name": "Jeffrey", "Last_Name": "Ullman" },
18                  { "First_Name": "Jennifer", "Last_Name": "Widom" } ]
19    }
20  ],
21  "Magazines":
22  [
23    { "Title": "New York Times",
24      "Price": 100,
25      "Remark": "Buy this book bundled with 'A First Course' - a
26 great deal!",
27      "Authors": [ { "First_Name": "Hector", "Last_Name": "Garcia
28 -Molina" },
29                  { "First_Name": "Jeffrey", "Last_Name": "Ullman" },
30                  { "First_Name": "Jennifer", "Last_Name": "Widom" } ]
31    }
32  ]
33 }

```

Error?

```

1 { "Books":
2   [
3     { "ISBN":"ISBN-0-13-713526-2",
4       "Price":"85",
5       "Edition":nil,
6       "Title":"A First Course in Database Systems",
7       "Authors":[ {"First_Name":"Jeffrey", "Last_Name":"Ullman"},
8                   {"First_Name":"Jennifer", "Last_Name":"Widom"} ]
9     },
10    { "ISBN":"ISBN-0-13-815504-6",
11      "Price":100,
12      "Remark":"Buy this book bundled with 'A First Course' - a
13      great deal!",
14      "Title":"Database Systems:The Complete Book",
15      "Authors":[ {"First_Name":"Hector", "Last_Name":"Garcia
16                  -Molina"},
17                  {"First_Name":"Jeffrey", "Last_Name":"Ullman"},
18                  {"First_Name":"Jennifer", "Last_Name":"Widom"} ]
19    }
20  ]
21 }
```

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**NOT A RESERVED WORD,
NO QUOTES**

**ARRAYS DON'T NEED TO
HAVE THE SAME TYPE OF
VALUES AS THEIR ELEMENTS**

First element of Authors changed to simple string



powered by ace

```
1
2 { "Books":
3   [
4     { "ISBN":"ISBN-0-13-713526-2",
5       "Price":85,
6       "Edition":3,
7       "Title":"A First Course in Database Systems",
8       "Authors":["Jeffrey Ullman",
9                  {"First_Name":"Jennifer", "Last_Name":"Widom"} ]
10    },
11    { "ISBN":"ISBN-0-13-815504-6",
12      "Price":100,
13      "Remark":"Buy this book bundled with 'A First Course' - a
14      great deal!",
15      "Title":"Database Systems:The Complete Book",
16      "Authors":[{"First_Name":"Hector", "Last_Name":"Garcia
17      -Molina"},
18                 {"First_Name":"Jeffrey", "Last_Name":"Ullman"},
19                 {"First_Name":"Jennifer", "Last_Name":"Widom"} ]
20    }
21  ],
22  "Magazines":
23  [
```

Empty elements also acceptable (see Magazines)

```
5      "Price":85,
6      "Edition":3,
7      "Title":"A First Course in Database Systems",
8      "Authors":[ {"First_Name":"Jeffrey", "Last_Name":"Ullman"},
9                  {"First_Name":"Jennifer", "Last_Name":"Widom"} ]
10     },
11     { "ISBN":"ISBN-0-13-815504-6",
12       "Price":100,
13       "Remark":"Buy this book bundled with 'A First Course' - a
14       great deal!",
15       "Title":"Database Systems:The Complete Book",
16       "Authors":[ {"First_Name":"Hector", "Last_Name":"Garcia
17                   -Molina"},
18                   {"First_Name":"Jeffrey", "Last_Name":"Ullman"},
19                   {"First_Name":"Jennifer", "Last_Name":"Widom"} ]
20     }
21 ],
22 "Magazines":
23 [
24   { }
25   ,
26   { "Title":"Newsweek"
```

Array length can be zero (see Authors)

```
1
2 { "Books":
3   [
4     { "ISBN":"ISBN-0-13-713526-2",
5       "Price":85,
6       "Edition":3,
7       "Title":"A First Course in Database Systems",
8       "Authors":[ {"First_Name":"Jeffrey", "Last_Name":"Ullman"},
9                   {"First_Name":"Jennifer", "Last_Name":"Widom"} ] }
10    ,
11    { "ISBN":"ISBN-0-13-815504-6",
12      "Price":100,
13      "Remark":"Buy this book bundled with 'A First Course' - a great
14      deal!",
15      "Title":"Database Systems:The Complete Book",
16      "Authors":[ ] }
17    ],
18   "Magazines":
19   [
20     { }
21     ,
22     { "Title":"Newsweek",
23       "Month":"February",
24       "Year":2009 }
```

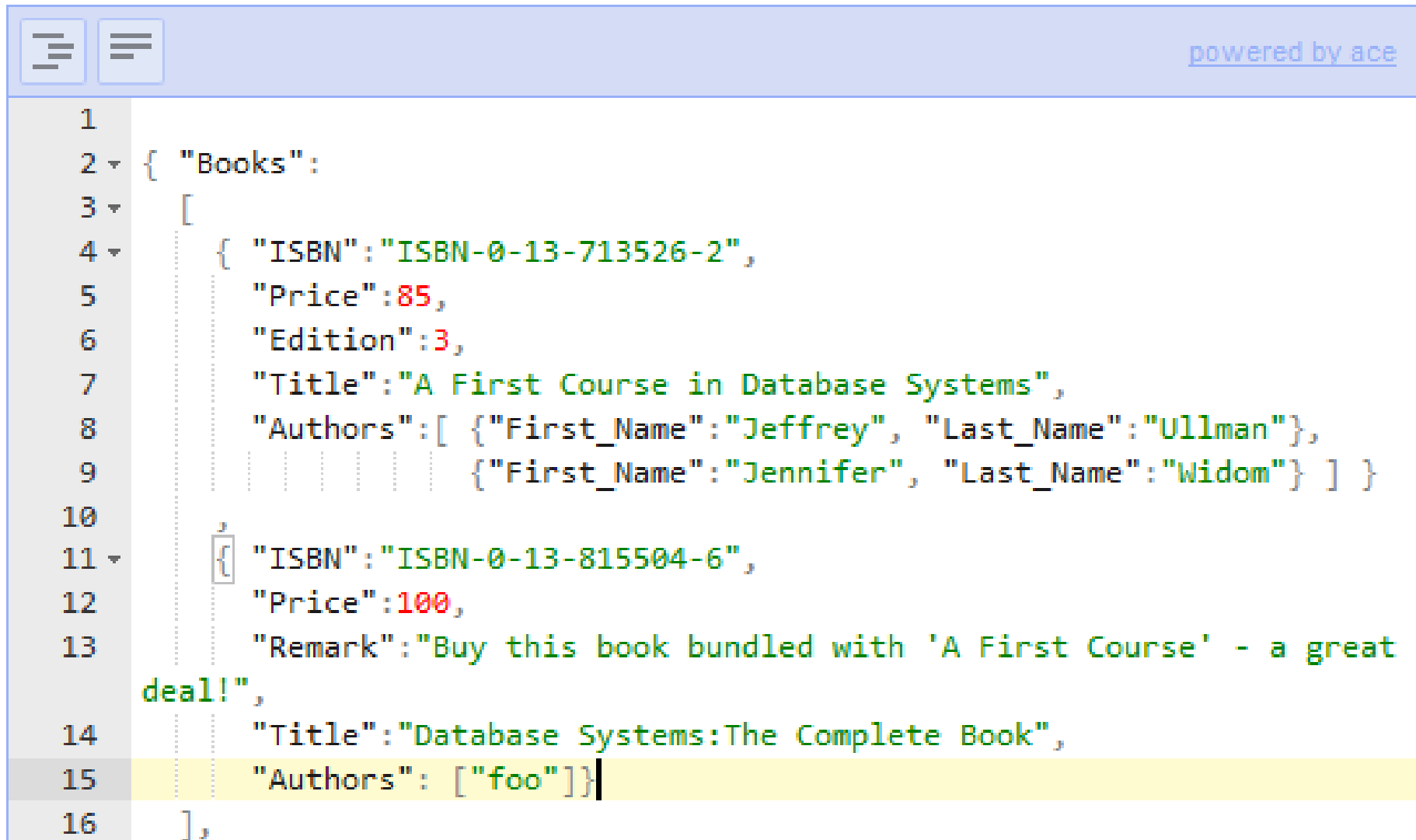


```
1
2 { "Books":
3   [
4     { "ISBN":"ISBN-0-13-713526-2",
5       "Price":85,
6       "Edition":3,
7       "Title":"A First Course in Database Systems",
8       "Authors":[ {"First_Name":"Jeffrey", "Last_Name":"Ullman"},
9                   {"First_Name":"Jennifer", "Last_Name":"Widom"} ] },
10    ,
11    { "ISBN":"ISBN-0-13-815504-6",
12      "Price":100,
13      "Remark":"Buy this book bundled with 'A First Course' - a great
14      deal!",
15      "Title":"Database Systems:The Complete Book",
16      "Authors": },
17    ],
18    "Magazines":
19    [
20      { },
21      { "Title":"Newsweek",
22        "Month":"February",
```

Error?

**ENTIRE VALUE IS NOW
MISSING FROM THE LABEL-
VALUE PAIR**

One string element in the array is acceptable



```
1
2 { "Books":
3   [
4     { "ISBN":"ISBN-0-13-713526-2",
5       "Price":85,
6       "Edition":3,
7       "Title":"A First Course in Database Systems",
8       "Authors":[ {"First_Name":"Jeffrey", "Last_Name":"Ullman"},
9                   {"First_Name":"Jennifer", "Last_Name":"Widom"} ] },
10
11     { "ISBN":"ISBN-0-13-815504-6",
12       "Price":100,
13       "Remark":"Buy this book bundled with 'A First Course' - a great
14       deal!",
15       "Title":"Database Systems:The Complete Book",
16       "Authors": ["foo"] }
```



```
1
2 { "Books":
3   [
4     { "ISBN":"ISBN-0-13-713526-2",
5       "Price":85,
6       "Edition":3,
7       "Title":"A First Course in Database Systems",
8       "Authors":[ {"First_Name":"Jeffrey", "Last_Name":"Ullman"},
9                   {"First_Name":"Jennifer", "Last_Name":"Widom"} ] }
10
11     { "ISBN":"ISBN-0-13-815504-6",
12       "Price":100,
13       "Remark":"Buy this book bundled with 'A First Course' - a great
14       deal!",
15       "Title":"Database Systems:The Complete Book",
16       "Authors": {"foo"}}
17   ],
18   "Magazines":
19   [
20     { }
21     ,
22     { "Title":"Newsweek",
23       "Month":"February",
24       "Year":2000 }
```

Error?

**THE VALUE IS AN OBJECT,
AND OBJECTS ARE ALWAYS
LABEL-VALUE PAIRS**

This is acceptable

```
1
2 { "Books":
3   [
4     { "ISBN":"ISBN-0-13-713526-2",
5       "Price":85,
6       "Edition":3,
7       "Title":"A First Course in Database Systems",
8       "Authors":[ {"First_Name":"Jeffrey", "Last_Name":"Ullman"},
9                   {"First_Name":"Jennifer", "Last_Name":"Widom"} ] },
10    { "ISBN":"ISBN-0-13-815504-6",
11      "Price":100,
12      "Remark":"Buy this book bundled with 'A First Course' - a great
13      deal!",
14      "Title":"Database Systems:The Complete Book",
15      "Authors": {"foo":"foo"}},
16  ],
17  "Magazines":
18  [
19    { }
20  ],
21  { "Title":"The Complete Book",
```

Examples Recap

- Syntactic correctness examples
- No required uniformity across values and arrays, or in objects

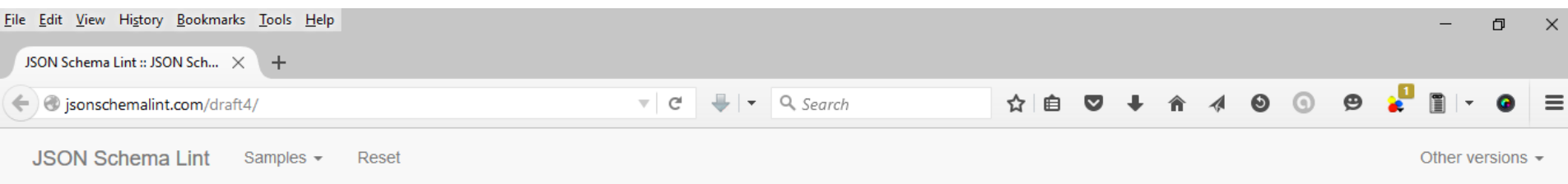
**JSON SCHEMA: ADD
CONSTRAINTS BEYOND
BASIC SYNTACTIC
CORRECTNESS**

**LIKE XML SCHEMA,
EXPRESSED IN THE DATA
MODEL ITSELF**

JSON SCHEMA

```
{ "type": "object",
  "properties": {
    "Books": {
      "type": "array",
      "items": {
        "type": "object",
        "properties": {
          "ISBN": { "type": "string", "pattern": "ISBN*" },
          "Price": { "type": "integer",
            "minimum": 0, "maximum": 200 },
          "Edition": { "type": "integer", "optional": true },
          "Remark": { "type": "string", "optional": true },
          "Title": { "type": "string" },
          "Authors": {
            "type": "array",
            "minItems": 1,
            "maxItems": 10,
            "items": {
              "type": "object",
              "properties": {
                "First_Name": { "type": "string" },
                "Last_Name": { "type": "string" } } } } } } },
    "Magazines": {
      "type": "array",
      "items": {
        "type": "object",
        "properties": {
          "Title": { "type": "string" },
          "Month": { "type": "string",
            "enum": [ "January", "February" ] },
          "Year": { "type": "integer" } } } }
  }
}
```

Validator



JSON Schema Lint is a JSON schema validator to help you write and test of [JSON Schemas](#) that conform with the Draft v4 specification.

The author/maintainer is [Nick Maynard](#). Fork [this project on Github](#).

Under the covers, it uses [Mathias Buus's is-my-json-valid](#) library, passed through [Browserify](#) to make it work in the browser.

Optionally, you may use schemas and documents in the YAML format. These documents are parsed with [Jérémy Fairve's yamli.js](#) library.

JSON Schema

Format

```
{
  "type": "object",
  "properties": {
    "Books": {
      "type": "array",
      "items": {
        "type": "object",
        "properties": {
          "ISBN": { "type": "string", "pattern": "ISBN*" },
          "Price": { "type": "integer",
            "minimum": 0, "maximum": 200 },
          "Edition": { "type": "integer", "optional": true },
          "Remark": { "type": "string", "optional": true },
          "Title": { "type": "string" },
          "Authors": {
            "type": "array",
            "minItems": 1,
            "maxItems": 10,
            "items": {
```

Schema is a valid JSON schema.

JSON Document

Format

```
{
  "Books": [
    {
      "ISBN": "ISBN-0-13-713526-2",
      "Price": 85,
      "Edition": 3,
      "Title": "A First Course in Database Systems",
      "Authors": [ { "First_Name": "Jeffrey", "Last_Name": "Ullman" },
                    { "First_Name": "Jennifer", "Last_Name": "Widom" } ]
    },
    {
      "ISBN": "ISBN-0-13-815504-6",
      "Price": 100,
      "Remark": "Buy this book bundled with 'A First Course' - a great deal!",
      "Title": "Database Systems: The Complete Book",
      "Authors": [ { "First_Name": "Hector", "Last_Name": "Garcia-Molina" },
                    { "First_Name": "Jeffrey", "Last_Name": "Ullman" },
                    { "First_Name": "Jennifer", "Last_Name": "Widom" } ]
    }
  ],
  "Magazines":
```

Document conforms to the JSON schema.

Catches syntactic errors (comma removed)

File Edit View History Bookmarks Tools Help

JSON Schema Lint :: JSON Sch... X +

jsonschemalint.com/draft4/ Search

JSON Schema Lint Samples Reset Other versions

The author/maintainer is [Nick Maynard](#). Fork [this project on Github](#).

Under the covers, it uses [Mathias Buus's is-my-json-valid](#) library, passed through [Browserify](#) to make it work in the browser.

Optionally, you may use schemas and documents in the YAML format. These documents are parsed with [Jérémy Fairre's yaml.js](#) library.

JSON Schema

Format

```
{ "type": "object",
  "properties": {
    "Books": {
      "type": "array",
      "items": {
        "type": "object",
        "properties": {
          "ISBN": { "type": "string", "pattern": "ISBN*" },
          "Price": { "type": "integer",
                     "minimum": 0, "maximum": 200 },
          "Edition": { "type": "integer", "optional": true },
          "Remark": { "type": "string", "optional": true },
          "Title": { "type": "string" },
          "Authors": {
            "type": "array",
            "minItems": 1,
            "maxItems": 10,
            "items": {
```

Schema is a valid JSON schema.

JSON Document

Format

```
{ "Books":
  [
    { "ISBN": "ISBN-0-13-713526-2",
      "Price": 85,
      "Edition": 3,
      "Title": "A First Course in Database Systems",
      "Authors": [ { "First_Name": "Jeffrey", "Last_Name": "Hillman",
                     { "First_Name": "Jennifer", "Last_Name": "Widom" } ] }
    ,
    { "ISBN": "ISBN-0-13-815504-6"
      "Price": 100,
      "Remark": "Buy this book bundled with 'A First Course' - a great deal!",
      "Title": "Database Systems: The Complete Book",
      "Authors": [ { "First_Name": "Hector", "Last_Name": "Garcia-Molina",
                     { "First_Name": "Jeffrey", "Last_Name": "Hillman",
                     { "First_Name": "Jennifer", "Last_Name": "Widom" } ] }
    ],
    "Magazines":
```

Field	Error	Value
	Document is invalid JSON. Try http://jsonlint.com to fix it.	

SEMANTIC CORRECTNESS DEMO

Price Specification

```
type:"object",
properties: {
  "Books": {
    "type":"array",
    "items": {
      "type":"object",
      "properties": {
        "ISBN": { "type":"string", "pattern":"ISBN*" },
        "Price": { "type":"integer",
          "minimum":0, "maximum":200 },
        "Edition": { "type":"integer", "optional": true },
        "Remark": { "type":"string", "optional": true },
        "Title": { "type":"string" },
        "Authors": {
          "type":"array",
          "minItems":1,
          "maxItems":10,
          "items": {
            "type":"object",
            "properties": {
              "First_Name": { "type":"string" },
              "Last_Name": { "type":"string" }
            }
          }
        }
      }
    }
  },
  "Magazines": {
```

Price as a string will not validate

JSON Document

```
{ "Books":  
  [  
    { "ISBN":"ISBN-0-13-713526-2",  
      "Price":85,  
      "Edition":3,  
      "Title":"A First Course in Database Systems",  
      "Authors":[ { "First_Name":"Jeffrey", "Last_Name":"Ullman"},  
                   { "First_Name":"Jennifer", "Last_Name":"Widom"} ] },  
    '  
    { "ISBN":"ISBN-0-13-815504-6",  
      "Price":"100",  
      "Remark":"Buy this book bundled with 'A First Course' - a great deal!",  
      "Title":"Database Systems:The Complete Book",  
      "Authors":[ { "First_Name":"Hector", "Last_Name":"Garcia-Molina"},  
                   { "First_Name":"Jeffrey", "Last_Name":"Ullman"},  
                   { "First_Name":"Jennifer", "Last_Name":"Widom"} ] },  
  ],  
  "Magazines":
```

```
"ISBN": { "type":"string", "pattern":"ISBN"  
"Price": { "type":"integer",  
           "minimum":0, "maximum":200 },  
"Edition": { "type":"integer", "optional":  
"Remark": { "type":"string", "optional": t
```

Field	Error	Value
data.Books.*.Price	is the wrong type	"100"

Price above range will not validate

JSON Document

```
{ "Books":  
  [  
    { "ISBN":"ISBN-0-13-713526-2",  
      "Price":85,  
      "Edition":3,  
      "Title":"A First Course in Database Systems",  
      "Authors":[ { "First_Name":"Jeffrey", "Last_Name":"Ullman" },  
                   { "First_Name":"Jennifer", "Last_Name":"Widom" } ] },  
    ,  
    { "ISBN":"ISBN-0-13-815504-6",  
      "Price":300, <-----  
      "Remark":"Buy this book bundled with 'A First Course' - a great deal!",  
      "Title":"Database Systems:The Complete Book",  
      "Authors":[ { "First_Name":"Hector", "Last_Name":"Garcia-Molina"},  
                   { "First_Name":"Jeffrey", "Last_Name":"Ullman" },  
                   { "First_Name":"Jennifer", "Last_Name":"Widom" } ] }  
  ],  
  "Magazines":
```

```
"ISBN": { "type":"string", "pattern":"ISBN"  
"Price": { "type":"integer",  
           "minimum":0, "maximum":200 },  
"Edition": { "type":"integer", "optional":  
"Remark": { "type":"string", "optional": t
```

Field	Error	Value
data.Books.*.Price	is more than maximum	300

Pattern matching for strings

```
{ "type": "object",  
  "properties": {  
    "Books": {  
      "type": "array",  
      "items": {  
        "type": "object",  
        "properties": {  
          "ISBN": { "type": "string", "pattern": "ISBN*" },  
          "Price": { "type": "integer",  
                     "minimum": 0, "maximum": 200 },  
          "Edition": { "type": "integer", "optional": true },  
          "Remark": { "type": "string", "optional": true },  
          "Title": { "type": "string" },  
          "Authors": {  
            "type": "array",  
            "minItems": 1,  
            "maxItems": 10,  
            "items": {  
              "type": "object",  
              "properties": {
```

String should start with ISBN and can be followed by anything else.

ISBN*

Drop I from ISBN, validation fails

JSON Document

Format

```
{ "Books":  
  [  
    { "ISBN":"SBN-0-13-713526-2",  
      "Price":85,  
      "Edition":3,  
      "Title":"A First Course in Database Systems",  
      "Authors":[ {"First_Name":"Jeffrey", "Last_Name":"Ullman"},  
                   {"First_Name":"Jennifer", "Last_Name":"Widom"} ] },  
    ,  
    { "ISBN":"ISBN-0-13-815504-6",  
      "Price":100,  
      "Remark":"Buy this book bundled with 'A First Course' - a great deal!",  
      "Title":"Database Systems:The Complete Book",  
      "Authors":[ {"First_Name":"Hector", "Last_Name":"Garcia-Molina"},  
                   {"First_Name":"Jeffrey", "Last_Name":"Ullman"},  
                   {"First_Name":"Jennifer", "Last_Name":"Widom"} ] },  
  ],  
  "Magazines":
```

Field	Error	Value
data.Books.*.ISBN	pattern mismatch	"SBN-0-13-713526-2"

**CONSTRAIN NUMBER OF
ELEMENTS IN AN ARRAY**

Array Constraints

```
type:"object",
properties: {
  "Books": {
    "type":"array",
    "items": {
      "type":"object",
      "properties": {
        "ISBN": { "type":"string", "pattern":"ISBN*" },
        "Price": { "type":"integer",
          "minimum":0, "maximum":200 },
        "Edition": { "type":"integer", "optional": true },
        "Remark": { "type":"string", "optional": true },
        "Title": { "type":"string" },
        "Authors": {
          "type":"array",
          "minItems":1,
          "maxItems":10,
          "items": {
            "type":"object"
            properties: {
              "First_Name": { "type":"string" },
              "Last_Name": { "type":"string" }}}}}}}},
  "Magazines": {
    .. ..
```

Error on removing all authors

JSON Document

Format

```
{ "Books":  
  [  
    { "ISBN":"ISBN-0-13-713526-2",  
      "Price":85,  
      "Edition":3,  
      "Title":"A First Course in Database Systems",  
      "Authors":[ ] }  
    ,  
    { "ISBN":"ISBN-0-13-815504-6",  
      "Price":100,  
      "Remark":"Buy this book bundled with 'A First Course' - a great deal!",  
      "Title":"Database Systems:The Complete Book",  
      "Authors":[ {"First_Name":"Hector", "Last_Name":"Garcia-Molina"},  
                   {"First_Name":"Jeffrey", "Last_Name":"Ullman"},  
                   {"First_Name":"Jennifer", "Last_Name":"Widom"} ] }  
  ],  
  "Magazines":  
  [  

```

Field	Error	Value
data.Books.*.Authors	has less items than allowed	[]

Validates when minItem set to zero

JSON Schema

```
{ "type": "object",
  "properties": {
    "Books": {
      "type": "array",
      "items": {
        "type": "object",
        "properties": {
          "ISBN": { "type": "string", "minLength": 10 },
          "Price": { "type": "integer", "minimum": 0 },
          "Edition": { "type": "integer", "minimum": 1 },
          "Remark": { "type": "string", "minLength": 10 },
          "Title": { "type": "string", "minLength": 10 },
          "Authors": {
            "type": "array",
            "minItems": 0,
            "maxItems": 10,
            "items": {
```

Schema is a valid JSON schema.

JSON Document

Format

```
{ "Books":
  [
    { "ISBN": "ISBN-0-13-713526-2",
      "Price": 85,
      "Edition": 3,
      "Title": "A First Course in Database Systems",
      "Authors": [ ] }
    ,
    { "ISBN": "ISBN-0-13-815504-6",
      "Price": 100,
      "Remark": "Buy this book bundled with 'A First Course' - a great deal!",
      "Title": "Database Systems: The Complete Book",
      "Authors": [ { "First_Name": "Hector", "Last_Name": "Garcia-Molina" },
                    { "First_Name": "Jeffrey", "Last_Name": "Ullman" },
                    { "First_Name": "Jennifer", "Last_Name": "Widom" } ] }
  ],
  "Magazines":
  [
```

Document conforms to the JSON schema.

Or just remove the minItem constraint – still valid

```
"Edition": { "type": "integer", "optional": true },
"Remark": { "type": "string", "optional": true },
"Title": { "type": "string" },
"Authors": {
  "type": "array",
  "maxItems": 10,
  "items": {
    "type": "object",
```

Schema is a valid JSON schema.

**ADDING ITEMS THAT
AREN'T IN THE SCHEMA**

Add "rating" – validates – even though it is not in the schema

JSON Document

Format

```
{ "Books":  
  [  
    { "ISBN":"ISBN-0-13-713526-2",  
      "Price":85,  
      "Rating":7,  
      "Edition":3,  
      "Title":"A First Course in Database Systems",  
      "Authors":[ ] }  
    ,  
    { "ISBN":"ISBN-0-13-815504-6",  
      "Price":100,  
      "Remark":"Buy this book bundled with 'A First Course' - a great deal!",  
      "Title":"Database Systems:The Complete Book",  
      "Authors":[ { "First_Name":"Hector", "Last_Name":"Garcia-Molina"},  
                   { "First_Name":"Jeffrey", "Last_Name":"Ullman"},  
                   { "First_Name":"Jennifer", "Last_Name":"Widom"} ] }  
  ],  
  "Magazines":
```

Document conforms to the JSON schema.

"additionalProperties":false

**ADDITIONS ARE ALLOWED
BY DEFAULT – BUT CAN BE
DISALLOWED BY SETTING A
FLAG IN THE SCHEMA**

JSON Schema

Format

```
{ "type": "object",
  "properties": {
    "Books": {
      "type": "array",
      "items": {
        "type": "object",
        "additionalProperties": false,
        "properties": {
          "ISBN": { "type": "string", "pattern": "ISBN*" },
          "Price": { "type": "number", "minimum": 0 },
          "Rating": { "type": "number", "minimum": 1, "maximum": 5 },
          "Edition": { "type": "number", "minimum": 1 },
          "Remark": { "type": "string" },
          "Title": { "type": "string" },
          "Authors": {
            "type": "array",
            "maxItems": 10,
            "items": {
              "type": "object",
              "properties": {
                "First_Name": { "type": "string" },
                "Last_Name": { "type": "string" }
              },
              "required": [ "First_Name", "Last_Name" ]
            }
          }
        }
      }
    }
  }
}
```

Schema is a valid JSON schema.

JSON Document

Format

```
{ "Books":
  [
    { "ISBN": "ISBN-0-13-713526-2",
      "Price": 85,
      "Rating": 7,
      "Edition": 3,
      "Title": "A First Course in Database Systems",
      "Authors": [ ] }
    ,
    { "ISBN": "ISBN-0-13-815504-6",
      "Price": 100,
      "Remark": "Buy this book bundled with 'A First Course' - a great deal!",
      "Title": "Database Systems: The Complete Book",
      "Authors": [ { "First_Name": "Hector", "Last_Name": "Garcia-Molina" },
                    { "First_Name": "Jeffrey", "Last_Name": "Ullman" },
                    { "First_Name": "Jennifer", "Last_Name": "Widom" } ] }
  ],
  "Magazines":
  [ ]
}
```

Field

Error

Value

data.Books.*

has additional properties

"data.Books[i].Rating"

JSON Schema

Format

```
{ "type": "object",
  "properties": {
    "Books": {
      "type": "array",
      "items": {
        "type": "object",
        "additionalProperties": true,
        "properties": {
          "ISBN": { "type": "string", "pattern": "ISBN*" },
          "Price": { "type": "integer",
            "minimum": 0, "maximum": 200 },
          "Edition": { "type": "integer", "optional": true },
          "Remark": { "type": "string" },
          "Title": { "type": "string" },
          "Authors": {
            "type": "array",
            "maxItems": 10,
            "items": {
```

Schema is a valid JSON schema.

Changed to true -
validates both for
default or true

JSON Document

Format

```
{ "Books":
  [
    { "ISBN": "ISBN-0-13-713526-2",
      "Price": 85,
      "Rating": 7,
      "Edition": 3,
      "Title": "A First Course in Database Systems",
      "Authors": [ ] }
    ,
    { "ISBN": "ISBN-0-13-815504-6",
      "Price": 100,
      "Remark": "Buy this book bundled with 'A First Course' - a great deal!",
      "Title": "Database Systems: The Complete Book",
      "Authors": [ {"First_Name": "Hector", "Last_Name": "Garcia-Molina"},
                    {"First_Name": "Jeffrey", "Last_Name": "Ullman"},
                    {"First_Name": "Jennifer", "Last_Name": "Widom"} ] }
  ],
  "Magazines":
```

Document conforms to the JSON schema.

Missing First_Name – will not validate

```
"Title": "Database Systems: The Complete Book",  
"Authors": [ { "First_Name": "Hector", "Last_Name": "Garcia-Molina" },  
              { "Last_Name": "Ullman" },  
              { "First_Name": "Jennifer", "Last_Name": "Widom" } ] }
```

```
"type": "string" },  
"Authors": {  
  "type": "array",  
  "minItems": 1,  
  "maxItems": 10,  
  "items": {  
    "type": "object",  
    "properties": {  
      "First_Name": { "type": "string" },  
      "Last_Name": { "type": "string" } } } } } },  
"names": {  
  "type": "array"
```

Unless optional label is set to true

```
"Authors": {  
  "type": "array",  
  "minItems": 1,  
  "maxItems": 10,  
  "items": {  
    "type": "object",  
    "properties": {  
      "First_Name": { "type": "string", "optional": "true" },  
      "Last_Name": { "type": "string" } } } } } },  
"Addresses": {  
  "type": "array",
```

Enum – March value does not validate

JSON Document

```
    "Authors": [ { "First_Name": "Hector",  
                  { "Last_Name": "Ullman",  
                  { "First_Name": "Jennifer"  
    ],  
    "Magazines":  
    [  
      { "Title": "National Geographic",  
        "Month": "January",  
        "Year": 2009,  
      },  
      { "Title": "Newsweek",  
        "Month": "March",  
        "Year": 2009 }  
    ]  
  }
```

```
"Magazines": {  
  "type": "array",  
  "items": {  
    "type": "object",  
    "properties": {  
      "Title": { "type": "string" },  
      "Month": { "type": "string",  
                 "enum": ["January", "February"] },  
      "Year": { "type": "integer" } } } }
```

Field	Error	Value
data.Magazines.*.Month	must be an enum value	"March"

Turn object to string – **does not validate now** – schema specifies array of objects in Author

```
"Remark":
  "Buy this book bundled with 'A First Course' - a great deal!",
"Title": "Database Systems: The Complete Book",
"Authors": [ { "First_Name": "Hector",
               "Last_Name": "Garcia-Molina" },
             { "First_Name": "Jeffrey Ullman",
               "Last_Name": "Jeffrey" },
             { "First_Name": "Jennifer", "Last_Name": "Widom" } ] }

magazines": {
  "Title": "Database Systems: The Complete Book",
  "Authors": {
```

```
"Authors": {
  "type": "array",
  "minItems": 1,
  "maxItems": 10,
  "items": {
    "type": "object",
    "properties": {
      "First_Name": { "type": "string" },
      "Last_Name": { "type": "string" }}}}}},
"es": {
  " " " "
```

**DIFFERENT TYPES CAN BE
ALLOWED BY
SPECIFICATION – E.G.,
ALLOW YEAR AS INT &
STRING**

Original file will not validate with the year as a string

JSON Document

Format

```
    "Title": "Database Systems: The Complete Book",
    "Authors": [ { "First_Name": "Hector", "Last_Name": "Garcia-Molina" },
                  { "First_Name": "Jeffrey", "Last_Name": "Ullman" },
                  { "First_Name": "Jennifer", "Last_Name": "Widom" } ] },
    "Magazines": [
      { "Title": "National Geographic",
        "Month": "January",
        "Year": 2009 },
      { "Title": "Newsweek",
        "Month": "February",
        "Year": "2009" }
    ]
  }
```

Field	Error	Value
data.Magazines.*.Year	is the wrong type	"2009"

Change specification to array

```
type: { type: "string",  
  "Month": { "type": "string",  
    "enum": ["January", "February"] },  
  "Year": { "type": ["integer", "string"]| } } }
```

JSON Document

Format

```
"Title": "Database Systems: The Complete Book",
"Authors": [ {"First_Name": "Hector", "Last_Name": "Garcia-Molina"},
              {"First_Name": "Jeffrey", "Last_Name": "Ullman"},
              {"First_Name": "Jennifer", "Last_Name": "Widom"} ],
"Magazines": [
  { "Title": "National Geographic",
    "Month": "January",
    "Year": 2009 },
  { "Title": "Newsweek",
    "Month": "February",
    "Year": "2009" }
]
```

Document conforms to the JSON schema

JSON Schema

Format

```
    "maxItems": 10,
    "items": {
      "type": "object",
      "properties": {
        "First_Name": { "type": "string" },
        "Last_Name": { "type": "string" } } } } } },
"Magazines": {
  "type": "array",
  "items": {
    "type": "object",
    "properties": {
      "Title": { "type": "string" },
      "Month": { "type": "string",
        "enum": ["January", "February"] },
      "Year": { "type": ["integer", "string"] } } } }
}
```

Schema is a valid JSON schema.

Validating for
both string and
int year now

**END OF DEMO. NOT
EXHAUSTIVE. JUST AN
EXAMPLE.**

Content adapted from
http://www.webmonkey.com/2010/02/get_started_with_json
and <http://www.w3schools.com/json/default.asp>